

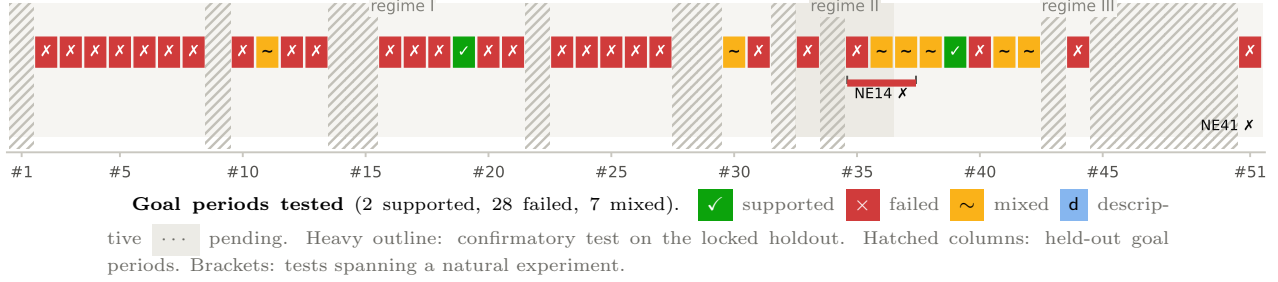
H42: Cross-excitation is a delayed step at the next read-out

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 62% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. Reading changes what the next call does, not when it comes: a named read raises talk at the read-out call by 0.076 [0.066, 0.086] (half of it outside any exchange), and this named kernel survives a fitted Cox field while H03's exponential cross term does not.

level: descriptive · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

H42 refits H03's multivariate Hawkes model with kernels anchored at the recipient's next model call (the read-out call). Round 1 (71 non-reserved units, 35 periods): talk locks to each agent's *own* call clock (+1.1 nats per held-out message, 57/57 units), and an ordinary read barely changes whether an agent talks ($n_{\text{cross}} \approx 0.004$ vs 0.061 for H03's kernel). Round 2 asks how reading acts. Call starts are mechanical (chains, timers, the chat schedule), so a read can act only through what the reading call does. In regime III a named read raises talk at that call by 0.076 per read. Half of this, 0.046, comes with no exchange in progress. The call clock moves only through this switch to talk. Under a fitted Cox field, H03's exponential term mostly vanishes (survival 0.22), while the named kernel keeps 0.90.

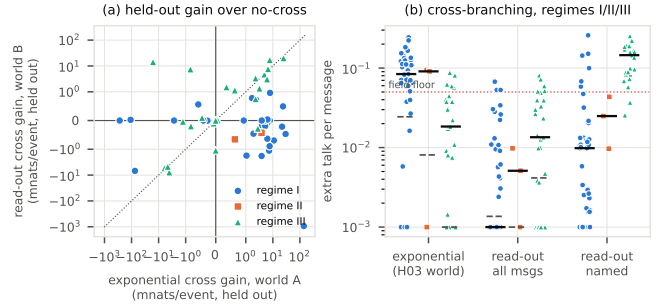
PHYSICAL MODEL

Two layers: the call skeleton (start times and classes) and the mark of each call. Agent i 's call c starts at t_c ; w_c is the smaller of its latency, the previous gap and 120 s. R_X^m counts messages of class X (cold named, thread named, unnamed) read at c and posted in $(t_c - w_c, t_c)$. P_X counts those posted in $(t_c, t_c + w_c)$, in flight. O_c is an outcome decided by call c : talk, timer pause, log gap to the next call, class of the next call. $a_{i,d,\kappa}$ is an agent $\times$ day $\times$ class cell. R3 replaces round 1's shared shape s_b with a free log-rate ϕ per day (and room) $\times$ 10-min bin.

$$O_c = a_{i,d,\kappa} + \sum_X [\beta_X^m R_{X,c}^m + \beta_X^P P_{X,c}] + \gamma \cdot E_c + \rho Y_{c-1}$$

$$J_{O,X} = \beta_X^m - \beta_X^P \quad (\text{minus the call-skeleton null})$$

$$\lambda_i^B = f_i(t - c_{k*}) \left[c_{id\kappa} e^{\phi_{d,r,b}} + \Phi_i^{\text{self}} + \sum_b \frac{w_b}{M_b} N_b(k^*) \right]$$



Observables. Round 1. (a) Per unit, held-out gain from cross-excitation: read-out kernel in the call-clock world vs exponential kernel in the H03 world (symlog). (b) Extra talk per message by regime (I, II, III): exponential, read-out over all messages, and read-out for named messages. Bars: medians; dashed: shift-null q95; red: synthetic shared-field floor.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Model each agent's call schedule and call class before you fit an influence model to agent logs. Then judge cross terms against a fitted common field, not shift surrogates. H03's exponential cross-triggering mostly disappears under such a field. To get a response, name the agent. In the always-on regime a named message adds 0.076 talk at the reader's next call, 0.046 even outside an ongoing exchange; an unnamed one adds 0.004. A message cannot start a call. It changes only what the next call does, so naming makes an agent speak sooner (about 3% shorter gap), not wake earlier.

RESULTS VS PREDICTIONS

FAITHFULNESS SCORECARD

Prediction (dated)	Observed	Verdict
P1 read-out beats exponential	gain > exponential in 33% (pre-reg. 89%, void)	failed
P3 exponential understates n_{cross}	0.004 vs 0.061	reversed
P5 step at calls 0–1	100% of mass at calls 0–1	supported
G1 synthetic guard	shared field passes every null	failed
<i>Round 2 (2026-10-05), regime III unless noted</i>		
R2-P1 named read cuts pausing	−0.0015 [−0.0033, 0.0003]	inconclusive
R2-P2 named read shortens gap	log gap −0.032 [−0.044, −0.019]; −0.009 n.s. given the mark	supported (fragile)
R2-P4 no class switch (I)	chat next −0.0017 unnamed, −0.0015 named	supported
R3-P1 exponential term is field	n_x 0.061 → 0.003; survival 0.22	supported
R3-P3 named term survives	0.142 → 0.108/msg; survival 0.90; held out 19/23	supported
R1-P1 agrees with H67	0.082 [0.076, 0.087] vs 0.079	supported
R1-P2 cold $\geq$ half of thread	0.046 vs 0.091 (ratio 0.51)	at threshold

A mapping 1 F identifiability 2
 B assumptions 1 G ground truth 1
 C adequacy 2 H comparative 2
 D unfitted predictions 1 I transfer 0
 E interventional 0
 Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

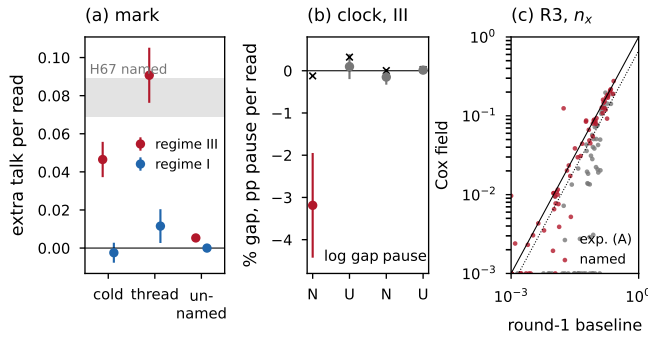
CAVEATS

The log-gap effect is fragile: a pooled synthetic field world gave 0.011, just above its 0.01 tolerance. With the mark held fixed it is not resolved, but that check conditions on a mediator (post hoc). The pause effect is underpowered. A 10-min Cox field absorbs about 25% of planted read-out excitation, so field survival understates real coupling. Thread status rests on DQ2 parents and recent naming, so some cold messages may sit in exchanges DQ2 missed. Regime-I log gap and unnamed session start fail the call-length world and are not read.

ROUND 2

Confirm the cold-named jump and the Cox-field named kernel on reserved data (re-freeze C5 with the thread split). Find who answers a cold name. Separate direct clock effects from the talk switch without conditioning on a mediator. Fit finer Cox fields with a smoothness prior.

- **Essence**. Whether, and through which messages, what an agent reads changes what it does next, measured on the recipient's call clock.
- **H42-R1** Named messages as the coupling carrier.
- **H42-R2** Two-layer point process.
- **H42-R3** Shared-field control.



Second figure. Round 2. (a) Talk jump per read at the read-out call (random-effects pools, 95% CI): cold-named, thread-named and unnamed messages; grey band: H67's named value. (b) Regime III, call timing per read, excess over the call-skeleton null (crosses: null mean): log gap to the next call (%) and timer-pause probability (pp), named (N) and unnamed (U). (c) Per unit, n_x with round 1's shared baseline vs a Cox field: H03's exponential term falls off the diagonal; the named kernel stays near it (dotted: 0.67, the synthetic survival of real coupling).

GOAL PERIODS IN DETAIL

G02 × failed (n_x read-out 0.022 vs exponential 0.166; held...
 G03 × failed (n_x read-out 0.067 vs exponential 0.141; held...
 G04 × failed (n_x read-out 0.030 vs exponential 0.121; held...
 G05 × failed (n_x read-out 0.026 vs exponential 0.087; held...
 G06 × failed (n_x read-out 0.000 vs exponential 0.129; held...
 G07 × failed (n_x read-out 0.000 vs exponential 0.126; held...
 G08 × failed (n_x read-out 0.000 vs exponential 0.016; held...
 G10 × failed (n_x read-out 0.010 vs exponential 0.078; held...
 G11 ~ mixed (n_x read-out 0.013 vs exponential 0.071; held...
 ...
 G12 × failed (n_x read-out 0.006 vs exponential 0.099; held...
 +27 more periods: see the card.